HAND AND OBJECT POSE TRACKING

In the MULTI-CAMERA SYSTEM

DR. YU XIANG | UTD – 16 | CAST STEM Bridge

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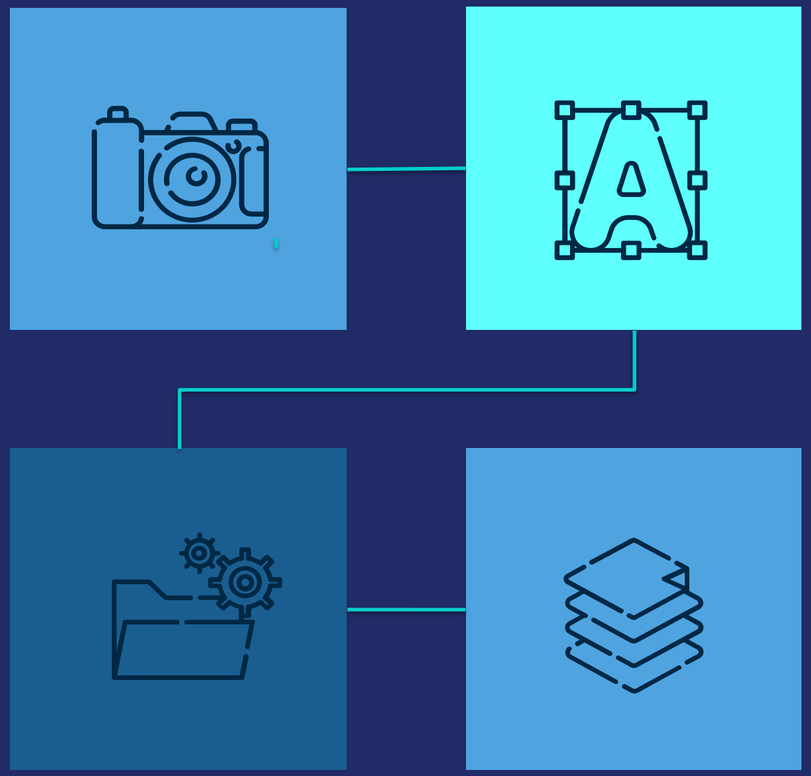
PROJECT DESCRIPTION

This project focuses on estimating hand and object poses from data collected using a multi-camera setup. The main objective is to develop an end-to-end pipeline for jointly tracking hand and object poses, which is crucial for computer vision applications commonly seen in robotics and automation, virtual and augmented reality (XR), etc.

PROJECT PIPELINE

Data Collection:
Hand-object manipulation recording, MANO hand shape calibration



Group A: Hand Pose Estimation
(MediaPipe, HaMeR)

Group B: Object Pose Estimation
(FoundationPose)

Combined Hand-object Poses from Group A & B's results

DATA COLLECTION

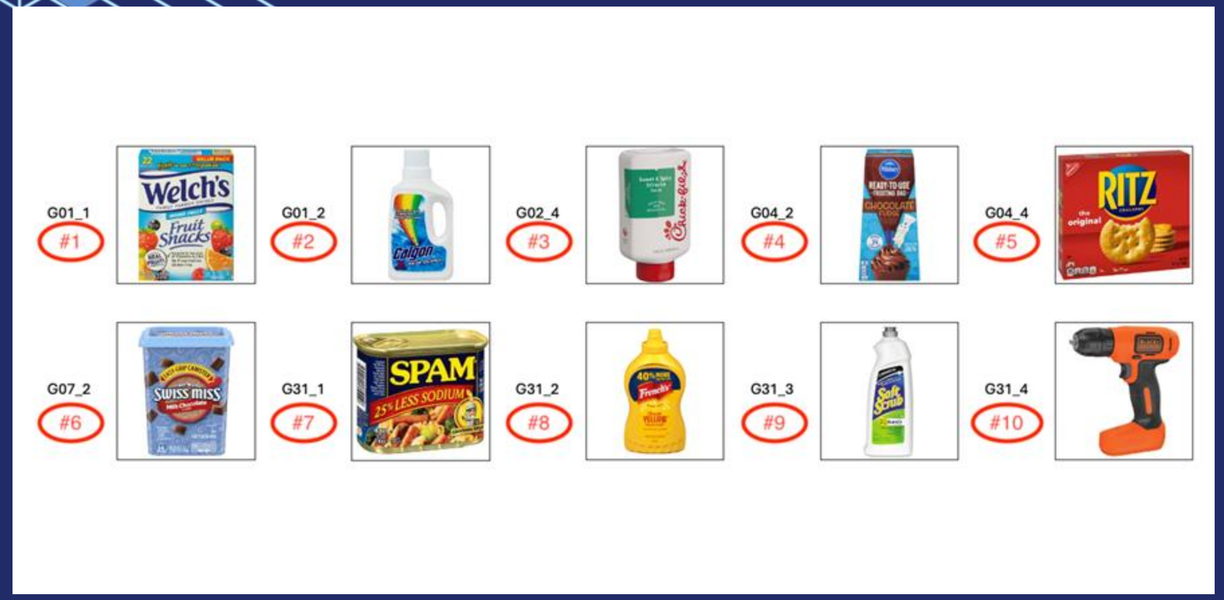
Multi-Camera System

Using multi-camera setup, we captured recordings of movement from eight angels

MANO Hand Shape Calibration

Fitting the MANO hand model to the merged point cloud collected by the multi-camera setup

DATA COLLECTION



Rosbag Recording

We use the ROS (Robot Operating System) to record and synchronize RGBD images from all cameras

Metadata Generation

Metadata is generated for each recording, including camera serials, object ids, person ids, etc.

GROUP A: Hand Pose Estimation

HaMeR

Input: Bounding boxes created from MediaPipe

Output: 2D landmarks predicted by HaMeR

Smoothness

Punish the pose jitters by smoothness loss over the sequence

MediaPipe

Use Mediapipe to get the **2D hand joint landmarks** in each camera view. And estimate 3D joints by triangulation and RANSAC algorithms

Optimization

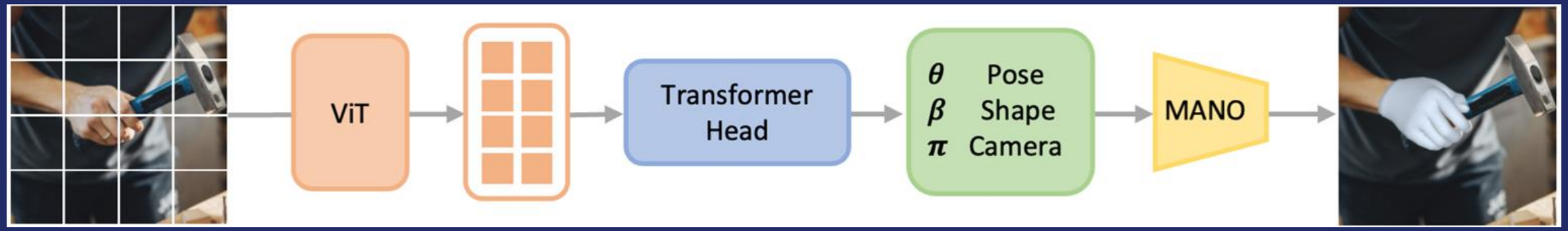
Optimize the MANO hand poses by 2D Projection Loss for each of the eight cameras

GROUP A: Hand Pose Estimation

HaMeR

HaMeR is an approach for Hand Mesh Recovery from a single image. Given a bounding box of the hand and the hand side (left/right), it uses a deep network to reconstruct the hand in 3D in the form of a MANO mesh.

GROUP A: Hand Pose Estimation Implementation

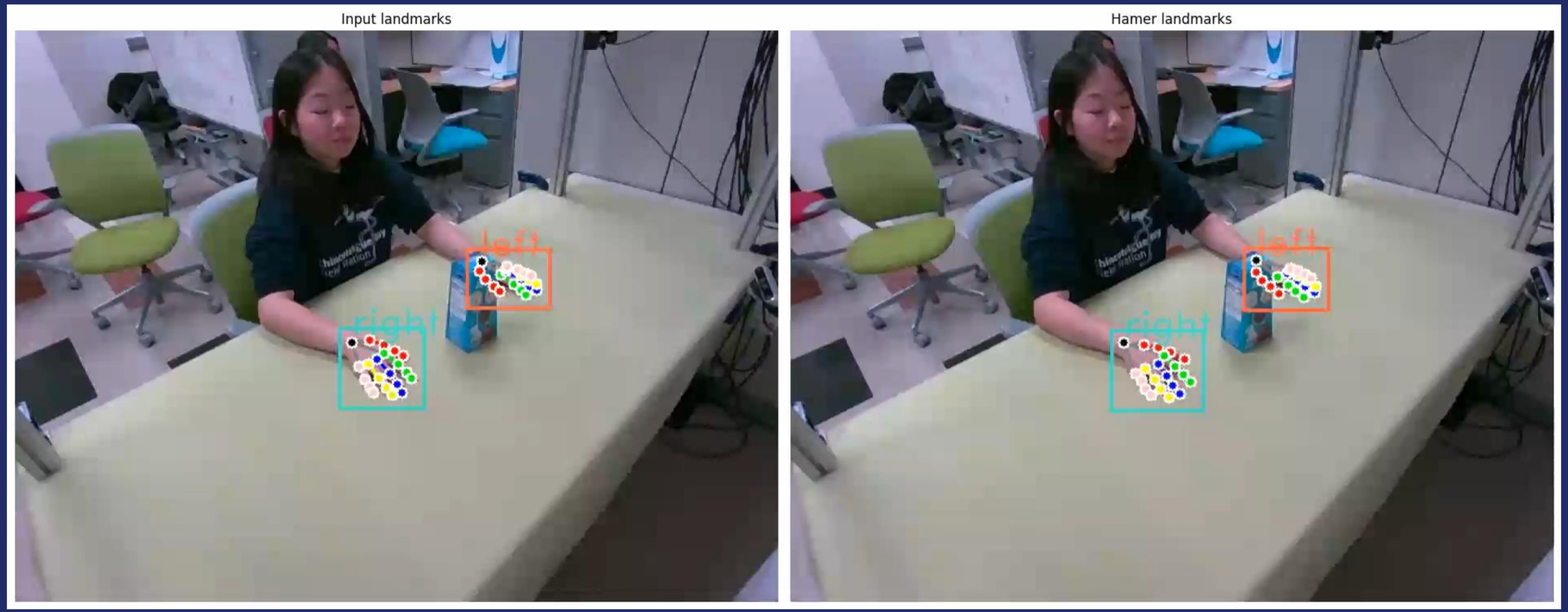


Implementing HaMeR Model

Initializing HaMeR model, loading data from recording (metadata, bounding box, landmarks), running inferences on the input data, and visualizing the output landmarks from the HaMeR model

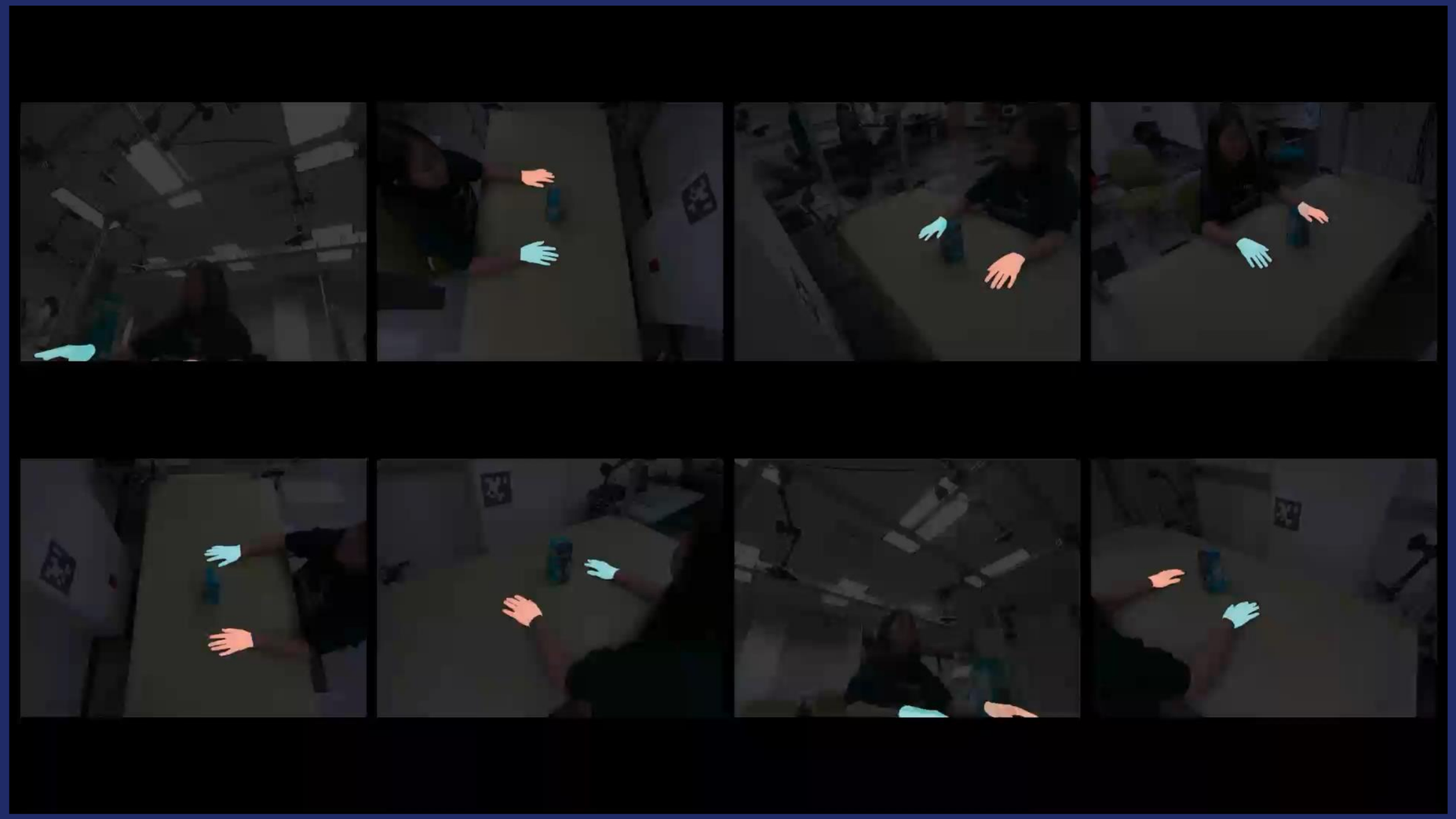
GROUP A: Hand Pose Estimation

MediaPipe & HaMeR Landmarks



GROUP A: Hand Pose Estimation

Optimized Hand Poses



GROUP B: Object Pose Estimation

Foundation Pose

Input: color, depth, mask, textured mesh
Output: the object pose in camera space

Smoothness

Punish the pose jitters by smoothness loss, and evaluate by visualizing rendered poses

Object Masks

Use **SAM** model to **segment the object mask** for the first frame, and **XMem** to get the remaining masks

Optimization

Estimate the best object pose in world space using RANSAC algorithm, with 6D pose loss in each camera

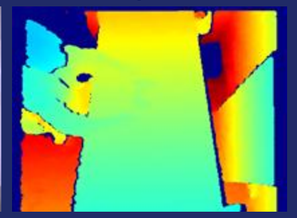
GROUP B: Object Pose Estimation Implementation

Input Data

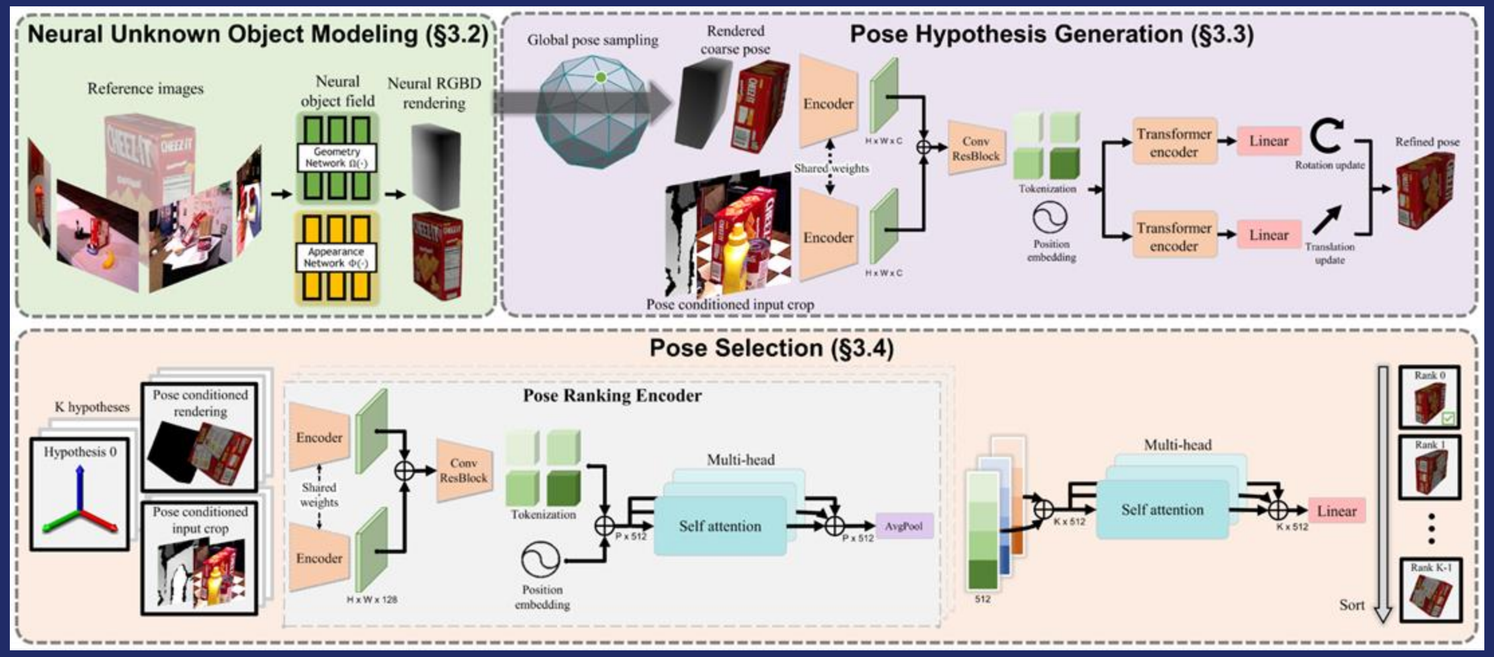
Textured 3D Mesh

Color

Depth



Mask



GROUP B: Object Pose Estimation

FoundationPose Result

GROUP B: Object Pose Estimation

Optimized Object Poses

Combined Hand-Object Poses

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THANK YOU!